

Solvation Thermodynamic Mapping of Molecular Surfaces in AmberTools: GIST

Steven Ramsey^{1,2}, Crystal Nguyen⁴, Romelia Salomon-Ferrer⁵, Ross C. Walker^{5,6}, Michael K. Gilson⁴, and Tom Kurtzman^{1,2,3}

Correspondence to: Tom Kurtzman (E-mail: thomas.kurtzman@lehman.cuny.edu), Mike Gilson (E-mail: mgilson@ucsd.edu), and Ross C. Walker (E-mail: ross@rosswalker.co.uk)

¹ *Lehman College Department of Chemistry, 205 w Bedford Pk Blvd Bronx New York, United States of America, 10468*

² *Ph.D. Program in Chemistry, The Graduate Center of the City University of New York, 365 5th Avenue, New York New York, United States of America, 10016*

³ *Ph.D. Program in Biochemistry, The Graduate Center of the City University of New York, 365 5th Avenue, New York New York, United States of America, 10016*

⁴ *Skaggs School of Pharmacy and Pharmaceutical Sciences, University of California San Diego, 9500 Gilman Drive, La Jolla, California, United States of America, 92093-0736*

⁵ *San Diego Supercomputer Center, University of California San Diego, 9500 Gilman Drive MC0505, La Jolla California, United States of America, 92093-0505*

⁶ *Department of Chemistry and Biochemistry, University of California San Diego, 9500 Gilman Drive La Jolla California, United States of America, 92093-0505*

Table S1. Quantities calculated and output by GIST, for a user-defined region.		
Quantity	Description	Units
Voxel	Voxel number	none
Xcoord	X coordinate of voxel	Å
Ycoord	Y coordinate of voxel	Å
Zcoord	Z coordinate of voxel	Å
Population	Mean number of waters in voxel	molecules
gO	Oxygen number density	bulk density of water oxygens
gH	Hydrogen number density	bulk density of water hydrogens
TS _{trans} -dens	Translational entropy	kcal/mol/Å ³
TS _{trans} -norm	Translational entropy	kcal/mol/
TS _{orient} -dens	Orientational entropy	kcal/mol/Å ³
TS _{orient} -norm	Orientational entropy	kcal/mol/
TS _{six} -dens	Total entropy	kcal/mol/Å ³
TS _{six} -norm	Total entropy	kcal/mol/
E _{ww} -dens	Water-water energy	kcal/mol/Å ³
E _{ww} -norm	Water-water energy	kcal/mol/
E _{sw} -dens	Solute-water energy	kcal/mol/Å ³
E _{sw} -norm	Solute-water energy	kcal/mol/
E _{ij} ^[a]	Voxel pair interaction energy	kcal/mol
Dipole-dens	Mean water dipole density	Debye/Å ³
Dipole x-dens	X component of water dipole density	Debye/Å ³
Dipole y-dens	Y component of water dipole density	Debye/Å ³
Dipole z-dens	Z component of water dipole density	Debye/Å ³
Neighbor-dens	Mean number of water neighbors ^[b] , normalized to water density	molecules/Å ³
Neighbor-norm	Mean number of water neighbors ^[b] , on per-water basis	molecules
Order ^[a]	Average tetrahedral order parameter	none
[a] Optional outputs. [b] Neighbor defined by O-O distance of less than 3.5Å.		

Table S2. Mean energy per water and number density for various water models.		
Water Model	Mean Energy (E _{ww} -norm kcal/mol)	Number density (1/Å ³)
TIP3P	-9.533	0.0329
TIP4PEW	-11.036	0.0332
TIP4P	-9.856	0.0332
TIP5P	-9.596	0.0329
TIP3PFW	-11.369	0.0334
SPCE	-11.123	0.0333
SPCFW	-11.873	0.0329

S1. GISTPP commands to perform end-states analysis on Factor Xa and ligand RRR.

First the perturbed volumes around FXA, RRR, and their complex must be defined:

```
gistpp -i gist-g0.dx -i2 RRR.pdb -op defbp -opt cutoff1  
10 -o perturbedRRR.dx
```

```
gistpp -i gist-g0.dx -i2 FXA.pdb -op defbp -opt cutoff1  
10 -o perturbedFXA.dx
```

```
gistpp -i gist-g0.dx -i2 FXARRR.pdb -op defbp -opt  
cutoff1 10 -o perturbedFXARRR.dx
```

Then, a GIST map of the number of waters in each voxel must be produced, alongside a map of total energy:

```
gistpp -i cpptraj.gist.out -i2 gist-g0.dx -op makedx  
-opt const 4 -o numwat.dx
```

```
gistpp -i gist-Esw-dens.dx -i2 gist-Eww-dens.dx -op mult  
-o gist-Etot-dens.dx
```

Next, we will multiply the perturbed volume maps by the created total energy map to quantify the total energy within the perturbed volumes:

```
gistpp -i gist-Etot-dens.dx -i2 perturbedRRR.dx -op mult  
-o pertRRREtot.dx
```

```
gistpp -i gist-Etot-dens.dx -i2 perturbedFXA.dx -op mult  
-o pertFXAEtot.dx
```

```
gistpp -i gist-Etot-dens.dx -i2 perturbedFXARRR.dx -op  
mult -o pertFXARRREtot.dx
```

Next, we will multiply the perturbed volume maps by the combined entropy map to quantify the total entropy within the perturbed volumes:

```
gistpp -i gist-TSsix-dens.dx -i2 perturbedRRR.dx -op  
mult -o pertRRRStot.dx
```

```
gistpp -i gist-TSsix-dens.dx -i2 perturbedFXA.dx -op  
mult -o pertFXAStot.dx
```

```
gistpp -i gist-TSsix-dens.dx -i2 perturbedFXARRR.dx -op  
mult -o pertFXARRRStot.dx
```

Next, we will multiply the perturbed volume maps by the map containing the number of waters created in the previous steps to quantify the number of waters within the perturbed volumes:

```
gistpp -i numwat.dx -i2 perturbedRRR.dx -op mult -o
pertRRR_N.dx
```

```
gistpp -i numwat.dx -i2 perturbedFXA.dx -op mult -o
pertFXA_N.dx
```

```
gistpp -i numwat.dx -i2 perturbedFXARRR.dx -op mult -o
pertFXARRR_N.dx
```

Then, the following calculations will produce the ΔG_{solv}^{FXA} , bulk values here are utilized for TIP3P water (values can be found above in Table S2):

$$\sum_{i=1}^i E_i^{FXA} = \text{gistpp -i pertFXAEtot.dx -op sum}$$

$$\sum_{i=1}^i S_i^{FXA} = \text{gistpp -i pertFXAStot.dx -op sum}$$

$$N^{FXA} = \text{gistpp -i pertFXA_N.dx -op sum}$$

$$\Delta E_{solv}^{FXA} = \left(\sum_{i=1}^i E_i^{FXA} \right) - N^{FXA}(-9.533)$$

$$\Delta S_{solv}^{FXA} = \left(\sum_{i=1}^i S_i^{FXA} \right) - N^{FXA}(0.000)$$

$$\Delta G_{solv}^{FXA} = \Delta E_{solv}^{FXA} - T \Delta S_{solv}^{FXA}$$

The same process is repeated for the ligand and complex. Finally when all three free energy terms have been described using the above GISTPP commands we can estimate ΔG_{solv} :

$$\Delta G_{solv} = \Delta G_{solv}^{comp} - (\Delta G_{solv}^{FXA} + \Delta G_{solv}^{RRR})$$